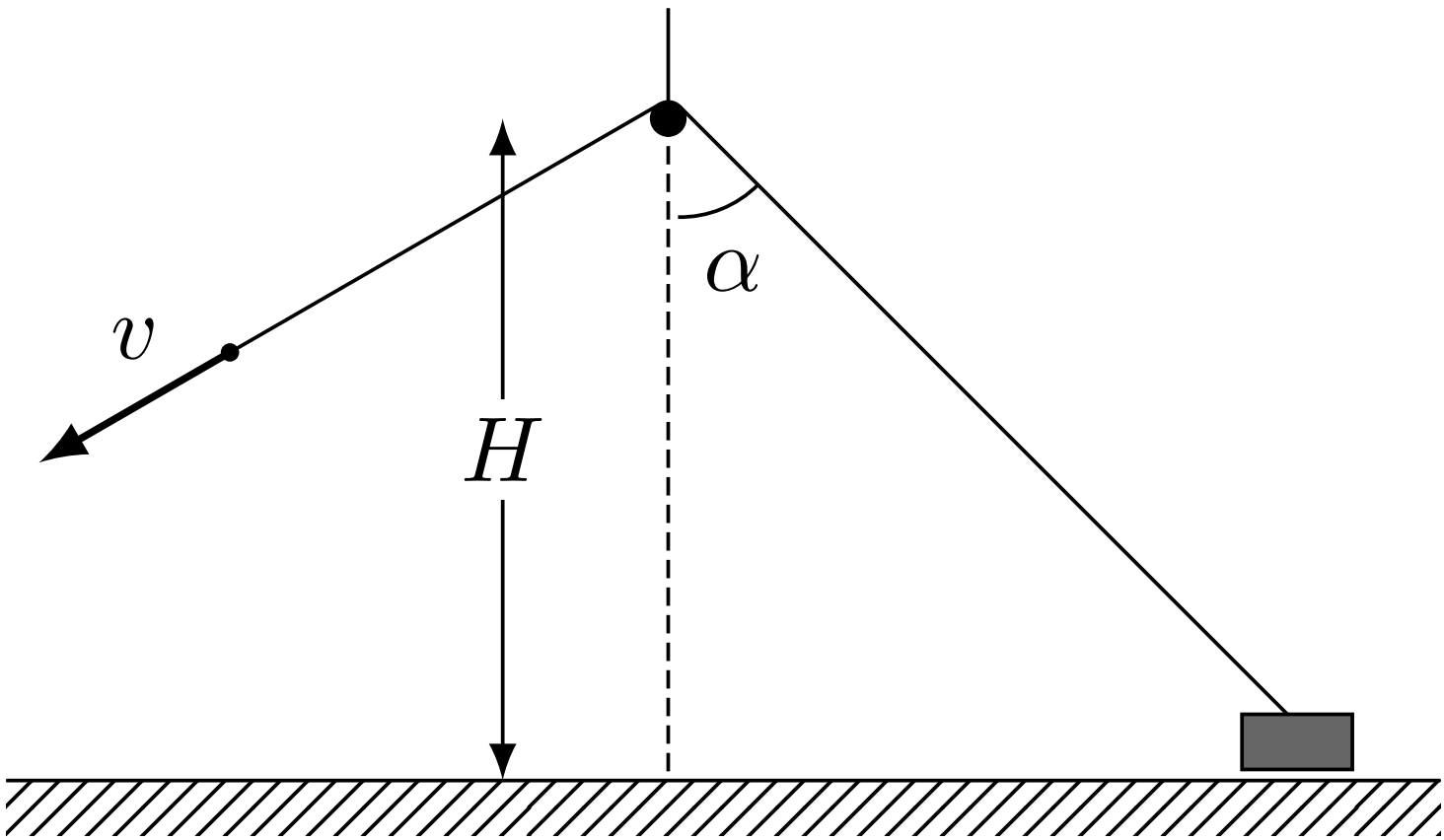
6. Pull-up bar rope (5 points) — *Jaan Kalda.* A rope is thrown over a horizontal pull-up bar at height H above the frictionless floor. One end of the rope is tied to a heavy weight resting on the floor; the other

end is pulled so that the length of the rope between the bar and the weight decreases at a constant rate v. At a given moment, this segment makes an angle α with the vertical (see figure); the weight is still in contact with the floor. Gravitational acceleration is g.



i) *(1 point)* Find the speed u of the weight.

ii) *(1 point)* Find the magnitude of the acceleration of the weight.

iii) *(3 points)* Find the angle α_0 at which the weight lifts off the floor.